

Operations Research in the Real World

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June 26, 2025

Today's objective:

- Identify at least one unintended consequence of their modeling choices.
- Identify the trade-offs between different decisions and strategies.
- Offer at least one alternative modeling strategy.

Resource Allocation Challenge: Northeast Unity Foundation

Northeast Unity Foundation faces critical decisions distributing limited resources among domestic violence survivors. Insufficient funding constraints comprehensive service delivery while demand escalates dramatically across metropolitan regions.

The organization must strategically allocate personnel, emergency housing, legal advocacy, and therapeutic services among diverse populations. Current challenges include inadequate shelter capacity, overwhelming counselor caseloads, and limited legal representation availability.

Resource Allocation Challenge: Northeast Unity Foundation

Budget constraints force difficult prioritization between prevention programs and crisis intervention services. Multiple shelter facilities require specialized staffing and security protocols while emergency hotlines demand round-the-clock coverage.

Childcare provisions, job training workshops, and trauma therapy compete for limited space and professional expertise. Administrative overhead reduces direct service funding, creating tension between operational efficiency and program expansion.

Resource Allocation Challenge: Northeast Unity Foundation

Grant restrictions limit flexible resource deployment, preventing responsive adaptation to emerging needs. Transportation assistance, technology resources, and multilingual services require additional allocations amid competing priorities.

Strategic decisions must balance immediate crisis response with long-term empowerment initiatives, ensuring sustainable support systems while maximizing impact across vulnerable populations.

Claude.ai created this case study
based on my research experience

Understanding the possible models

Model 1: Deterministic

Use a Deep Learning model to assess parameters based on data, and use deterministic model to optimize it.

DM is the NGO with binary decision in resource allocation.

Model 2: Stochastic

Stochasticity in:

- Parameters capacity
- Knowledge of decision-makers

DM is the Funding Organization. They have incomplete information on resource allocation' outcomes

Model 3: Multi-Criteria

Multi-criteria model considers decision of

- Funding orgs.
- Population served in 3 main groups.

DM is centralized, but criteria is weighted.

Activity - Discuss with your groups:

- Put in order of preferences the 3 choices.
- Make it clear why you are choosing one over the other.
- The why you are not choosing the model is also relevant.

Group 1 slide - vote here

Model 1: Deterministic

Model 2: Stochastic

Model 3: Multi-Criteria

1st Place

2nd Place

3rd Place

Group 2 slide - vote here

Model 1: Deterministic

Model 2: Stochastic

Model 3: Multi-Criteria

1st Place

2nd Place

3rd Place

Group 3 slide - vote here

Model 1: Deterministic

Model 2: Stochastic

Model 3: Multi-Criteria

1st Place

2nd Place

3rd Place

Group 4 slide - vote here

Model 1: Deterministic

Model 2: Stochastic

Model 3: Multi-Criteria

1st Place

2nd Place

3rd Place

Activity - Discuss with your groups:

- What are the main limitations of the models?
- What are the consequences of their decision in the real-world?
 - Who is affected by it?
 - How are they affected?

What happens after you choose?

Questions to ask your group:

- Who is benefiting the most?
- What are the main limitations?
- Whose decisions are not being modeled?
- Who might be affected by the decision but have no voice in this model?

Do you want to change your choices?

- Rethink if your group want to change the order of preference.
- Decide “why” and “why nots”

Round 2: Group 1 slide - vote here

Model 1: Deterministic

Model 2: Stochastic

Model 3: Multi-Criteria

1st Place

2nd Place

3rd Place

Round 2: Group 2 slide - vote here

Model 1: Deterministic

Model 2: Stochastic

Model 3: Multi-Criteria

1st Place

2nd Place

3rd Place

Round 2: Group 3 slide - vote here

Model 1: Deterministic

Model 2: Stochastic

Model 3: Multi-Criteria

1st Place

2nd Place

3rd Place

Round 2: Group 4 slide - vote here

Model 1: Deterministic

Model 2: Stochastic

Model 3: Multi-Criteria

1st Place

2nd Place

3rd Place

Main Limitations

Solution 1:
Deterministic

-
-
-

Solution 2:
Stochastic

-
-
-

Solution 1:
Multi-criteria

-
-
-

Potential Consequences

Solution 1:
Deterministic

-
-
-

Solution 2:
Stochastic

-
-
-

Solution 1:
Multi-criteria

-
-
-

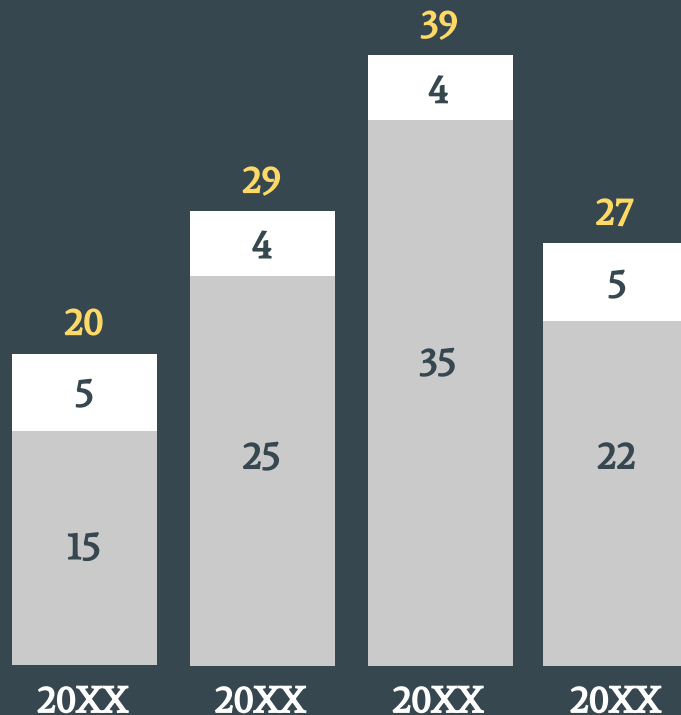
Overview of the activity

Findings

- The preferences of the class in the ranking system before and after the discussion.

Reflection

- What have you learned from this activity?
- What is the thought process to make this type of decision?



**What would you do
differently?**